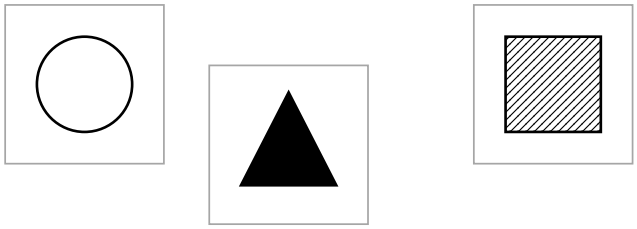
Use the nine tiles, one of each shape and fill. An allowed three uses three different tiles. Its shapes are all the same or all different. Its fills are also all the same or all different. Tile positions do not matter.



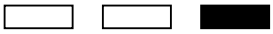
allowed three

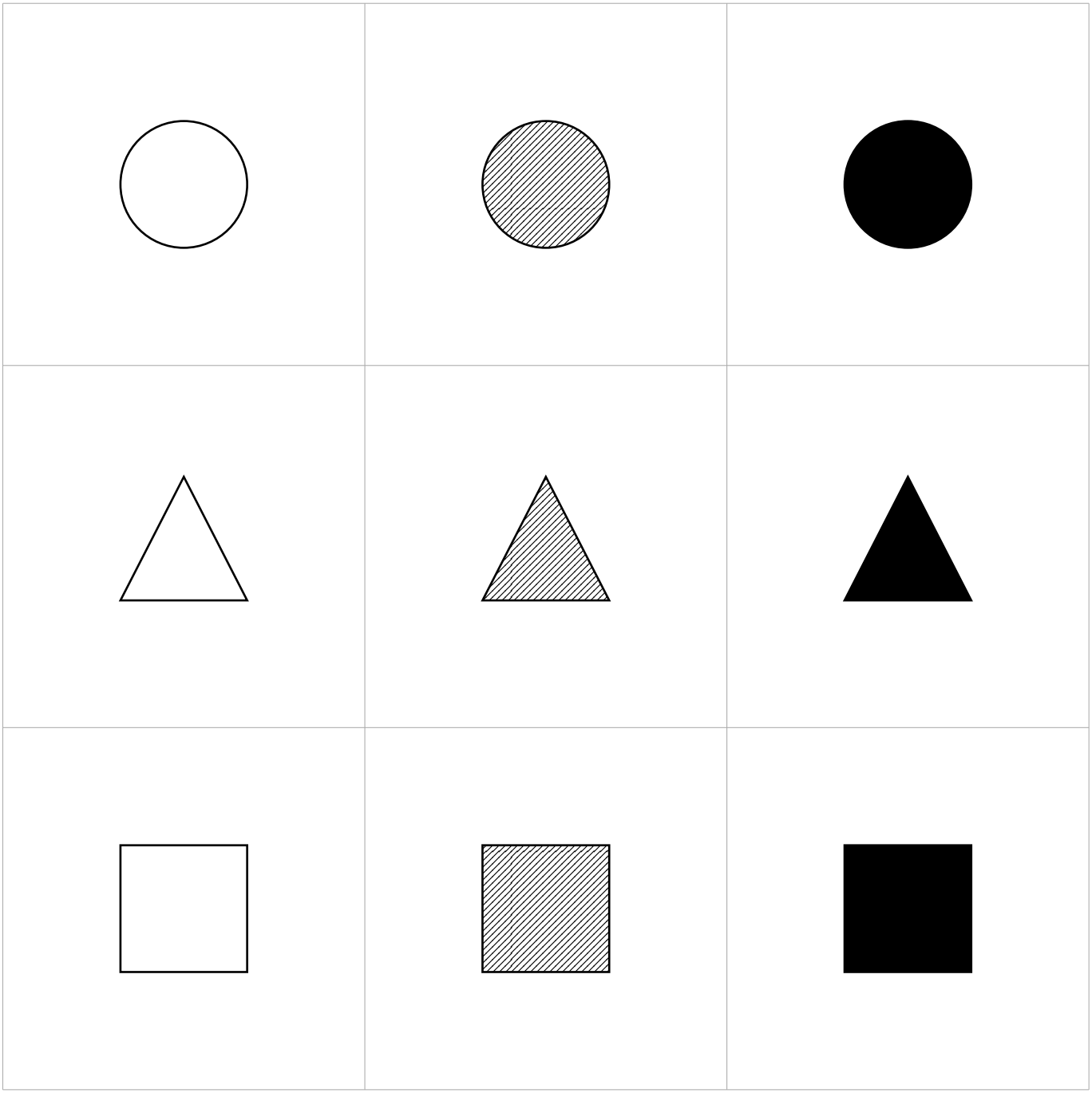
shapes all different

fills all different

not allowed

shapes all different

fills  two the same

Problem 1: Complete the pairs. Can two different tiles have two completing tiles, or none? Explain.

Problem 2: Choose as many tiles as possible without an allowed three among them. Explain why a larger collection is impossible.